(Please write your Enrollment Number) Enrollment No. 04801172022

MID-TERM EXAMINATION

(Programme: B. Tech. CSE-AI & ECE-AI) (Semester: VIII)

(March 2026) OFF LINE mode

Subject Code: BAI 416	**Subject: AI in Healthcare**
Time : 1 ½Hours	**Maximum Marks :30**

Note: Q1 is compulsory. Attempt any two parts of Q2 and Q3

Q1		**(2.5*4=10)**	**CO Mapping**
	a) What is time-series medical data? Give one example.		1
	b) Define data enrichment. Give suitable examples.		2
	c) Explain data sourcing and challenges in healthcare data collection.		2
	d) Define Clinical Decision Support System (CDSS).		3
	UNIT I		**CO Mapping**
Q2	**Attempt any two parts**	**(5*2=10)**	
	a) Explain the historical evolution of AI in healthcare with key milestones.		1
	b) A patient's respiratory rate (breaths/min) recorded every hour: *18, 20, 22, 24, 26, 30, 28* Questions: 1) Compute 3-point moving average. 2) Identify trend (increasing/decreasing). 3) Explain why moving average helps in time-series smoothing. 4) Mention one limitation of moving average in ICU monitoring.		4
	c) Analyze the impact of geographic and demographic variation on AI-based medical models.		3
	UNIT II		**CO Mapping**
Q3	**Attempt any two parts**	**(5*2=10)**	
	a) Explain different machine learning techniques used in disease detection using tabular medical data.		2
	b) Explain classification, regression, and clustering techniques in healthcare.		2
	c) Analyze challenges in implementing AI-driven CDSS in real clinical settings.		3